

Project Initialization and Planning Phase

Date	18 July 2026
Team ID	XXXXX
Project Title	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The Voice-Based Concept Understanding Analyser is an AI-enabled educational assessment system designed to evaluate a learner's conceptual understanding through spoken explanations. The application combines Whisper Speech Recognition for accurate speech-to-text conversion with Sentence-BERT semantic similarity to compare the learner's explanation against the reference concept. The system generates similarity scores, grammar analysis, AI-generated feedback, performance dashboards, downloadable PDF reports, and completion certificates. Developed using Python and Streamlit, the application provides a simple, interactive, and efficient platform for students and educators to improve learning outcomes through automated concept evaluation.

Project Overview	
Objective	The primary objective of the Voice-Based Concept Understanding Analyser project is to evaluate a learner's conceptual understanding through voice explanations using AI. The system aims to provide instant, accurate, and personalized feedback to help students improve their conceptual clarity, communication skills, and interview readiness.
Scope	The project includes voice recording or audio upload, speech-to-text conversion using Whisper AI, semantic similarity evaluation using Sentence-BERT, grammar and fluency analysis, AI-generated feedback, performance dashboards, PDF report generation, and certificate creation. The application is built using Python and Streamlit and is deployable online for students, educators, and institutions.
Problem Statement	
Description	Traditional assessments mainly focus on written answers, while oral evaluations are time-consuming, subjective, and provide delayed feedback. Students often struggle to identify weak concepts and communication gaps before exams or interviews. There is a lack of an intelligent system that can automatically evaluate spoken explanations and deliver immediate, objective, and meaningful feedback.
Impact	This solution enables learners to receive instant AI-driven feedback on their conceptual understanding and communication quality. It reduces the workload of educators, ensures fair and consistent evaluations, improves student learning outcomes, and strengthens interview and presentation preparation.
Proposed Solution	

Approach	The solution uses Whisper Speech Recognition to convert voice explanations into text and Sentence-BERT to evaluate the conceptual similarity between the learner's response and the reference concept. The Streamlit web application accepts voice recordings or audio uploads, processes the input, analyzes grammar and fluency, and generates similarity scores along with AI-based feedback. The system also provides performance dashboards, downloadable PDF reports, and completion certificates.
Key Features	Voice recording or audio file upload for concept explanation. Speech-to-text conversion using Whisper AI. Semantic similarity analysis using Sentence-BERT. Grammar and fluency evaluation of explanations. Instant AI feedback with similarity scores and suggestions. Interactive dashboards to track performance and progress. Downloadable PDF reports and completion certificates. Online deployment for easy and continuous access.
	- Personalized learning experience through AI-driven insights. - Scalable and efficient solution for students, educators, and institutions.

Resource Requirement

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	NVIDIA RTX 3050 / or Intel Core i5 (8th Gen or above)
Memory	RAM specifications	16 GB
Storage	Disk space for data, models, and logs	100 GB SSD
Software		
Frameworks	Python frameworks	Streamlit
Libraries	Additional libraries	whisper, sentence-transformers, reportlab, pandas, numpy, matplotlib, seaborn, mutagen
Development Environment	IDE	Visual Studio Code
Data		
Data	Source, size, format	Reference concept text files (.txt) User audio recordings (.wav/.mp3/.m4a) Generated reports (.pdf/.csv)

